

Math test -1Section - AQuestion - 1

(d) Reflexive, transitive but not symmetric

Question - 2

$$\text{Let } \theta = \sin^{-1}\left(-\frac{1}{2}\right)$$

$$\sin \theta = -\frac{1}{2}$$

$$\theta = -\frac{\pi}{6}$$

∴ Principal value is $\left(-\frac{\pi}{6}\right)$

Question - 3

$$A^2 = I$$

$$(A - I)^3 + (A + I)^3 - 7A$$

$$(A - A^2)^3 + (A + A^2)^3 - 7A$$

$$A^3 - 3A^2I + 3AI^2 - I^3 + A^3 + 3A^2I + 3AI^2 + I^3 - 7A$$

$$2A^3 + 6AI^2 - 7A$$

$$2A^3 + 6A$$

$$2A(A^2 + 3)$$

$$A = I$$

$$A^3 = A^2 \cdot A = A = I$$

$$2A(I)$$

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Question - 4

$$|\vec{a}| = \sqrt{2^2 + 3^2 + 1^2} = \sqrt{4+9+1} = \sqrt{14}$$

$$\hat{a} = \frac{\vec{a}}{|\vec{a}|} = \frac{2\hat{i} + 3\hat{j} + \hat{k}}{\sqrt{14}} = \frac{2}{\sqrt{14}}\hat{i} + \frac{3}{\sqrt{14}}\hat{j} + \frac{1}{\sqrt{14}}\hat{k}$$

$$\hat{a} = \frac{2}{\sqrt{14}}\hat{i} + \frac{3}{\sqrt{14}}\hat{j} + \frac{1}{\sqrt{14}}\hat{k}$$

Question - 5

$$P(A) = \frac{1}{2} \quad P(B) = 0$$

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

= undefined

(C) Not defined

Question - 6

$$y = \log(\sin(e^x))$$

$$\frac{dy}{dx} = \frac{1}{\sin(e^x)} \cdot \frac{d}{dx} [\sin(e^x)]$$

$$= \frac{1}{\sin(e^x)} \cdot \cos(e^x) \cdot \frac{d}{dx} (e^x)$$

$$= \cot(e^x) \cdot (e^x)$$

$$\therefore \int e^x \cot(e^x) dx$$

Question - 7

$$f(x) = 3 - 4x$$

One - one

$$x_1, x_2 \in \mathbb{R}$$

$$f(x_1) = f(x_2)$$

$$3 - 4x_1 = 3 - 4x_2$$

$$x_1 = x_2$$

∴ it is a one-one function

Onto

$$y = 3 - 4x$$

$$4x = 3 - y$$

$$x = \frac{3-y}{4}$$

∴ Since for every $y \in \mathbb{R}$, $x = \frac{3-y}{4} \in \mathbb{R}$ such that $f(x) = y$,
 f is also an onto function.

∴ f is bijective.

Question - 8

~~Question~~

$$\vec{a} = \hat{i} + 3\hat{j} + 7\hat{k}$$

$$\vec{b} = 7\hat{i} - \hat{j} + 8\hat{k}$$

$$\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$$

$$\vec{a} \cdot \vec{b} = (1 \times 7) + (3 \times -1) + (7 \times 8) = 7 - 3 + 56 = 60$$

$$|\vec{b}| = \sqrt{7^2 + (-1)^2 + (8)^2} = \sqrt{49 + 1 + 64} = \sqrt{114}$$

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$$\therefore \text{Projection} = \frac{60}{\sqrt{114}}$$

Question - 9

$$\text{Profit, } Z = 50A + 60B$$

$$5x + 10y \leq 180$$

Cutting: 3 hours = 180 minutes

Assembling: 4 hours = 240 minutes

$$\text{Subject to Constraints: } 5x + 8y \leq 180$$

$$10x + 8y \leq 240$$

Section-C

Question - 11

$$y = (\sin x)^x + \sin^{-1}(\sqrt{x})$$

$$\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$$

$$u = (\sin x)^x$$

$$\ln u = x \ln(\sin x)$$

$$v = \sin^{-1}(\sqrt{x})$$

$$\frac{dv}{dx} = \frac{1}{\sqrt{1-(\sqrt{x})^2}} = \frac{1}{\sqrt{1-x}}$$

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$$\frac{dv}{dx} = \frac{1}{\sqrt{1-(-x)^2}} \cdot \frac{d(-x)}{dx} = \frac{1}{\sqrt{1-x}} \cdot \frac{-1}{2\sqrt{x}}$$

$$= \frac{-1}{2\sqrt{x}(1-x)}$$

Question - 12

$$\frac{dy}{dx} = x \ln(8\pi x) + \frac{1}{2\sqrt{x}(1-x)}$$

Question - 12

$$f(x) = 2x^3 - 3x^2 - 36x + 7$$

Question - 13

Let E_1 = event that Bag 1 is chosen

E_2 = event that Bag 2 is chosen

E_3 = event that a red ball is drawn

$$P(E_1) = P(E_2) = \frac{1}{2}$$

$$P(A/E_1) = \frac{4}{8} = \frac{1}{2}$$

$$P(A/E_2) = \frac{2}{8} = \frac{1}{4}$$

By Baye's theorem:

$$P(E_1/A) = \frac{P(E_1) \cdot P(A/E_1)}{P(E_1) \cdot P(A/E_1) + P(E_2) \cdot P(A/E_2)}$$

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$$P(E_1/A) = \frac{\frac{1}{2} \times \frac{1}{2}}{\frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{4}}$$

$$= \frac{\frac{1}{4}}{\frac{1}{4} + \frac{1}{8}} = \frac{\frac{2}{8}}{\frac{3}{8}}$$

$$\therefore P(E_1/A) = \frac{2}{3} //$$

Section-D

~~Option-A~~

$$Z = -3x + 4y$$

Subject to:

$$x + 2y \leq 8$$

$$3x + 2y \leq 12$$

$$x \geq 0$$

$$y \geq 0$$

Option-B

Let the circle have radius R .

Let length & breadth be x & y respectively. Diagonal of a rectangle inscribed in a circle is equal to the diameter of the circle ($2R$):

$$x^2 + y^2 = (2R)^2 = 4R^2$$

$$y = \sqrt{4R^2 - x^2}$$

Let area of rectangle be A :

$$A = x \cdot y = x \cdot \sqrt{4R^2 - x^2}$$

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Consider $A^2 = X$

$$X = x^2(4R^2 - x^2) = 4R^2x^2 - x^4$$

Differentiating

$$\frac{dX}{dx} = 8R^2x - 4x^3$$

$$8R^2x - 4x^3 = 0$$

$$4x(2R^2 - x^2) = 0 \Rightarrow 2R^2 - x^2 = 0$$

Since $x > 0$,

$$x^2 = 2R^2$$

$$x = \sqrt{2}R$$

$$x = \sqrt{2}R$$

$$\frac{d^2X}{dx^2} = 8R^2 - 12x^2$$

$$= 8R^2 - 12(\sqrt{2}R)^2 = 8R^2 - 24R^2 = -16R^2 < 0$$

Since $\frac{d^2X}{dx^2} < 0$, $x = \sqrt{2}R$

$$y = \sqrt{4R^2 - x^2} = \sqrt{4R^2 - 2R^2} = \sqrt{2}R$$

$$y = \sqrt{2}R$$

Since $x = y = \sqrt{2}R$, the inscribed rectangle is a square.

Question - 10

$$f(x) = \begin{cases} \frac{k \cos x}{\pi - 2x}, & \text{if } x \neq \frac{\pi}{2} \\ 3, & \text{if } x = \frac{\pi}{2} \end{cases}$$

$$\lim_{x \rightarrow \frac{\pi}{2}} f(x) = f\left(\frac{\pi}{2}\right) = 3$$